Unveiling Fast-Mode Turbulence: Energy Spectrum & Cascade Dynamics.

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Abstract

Fast compressible modes in magnetohydrodynamic (MHD) turbulence are known to scatter cosmic rays much more efficiently than Alfvén modes, yet the dynamics behind their energy transfer remain under-explored. To address this gap, we conduct isothermal, 3D MHD simulations of decaying fast mode turbulence in *low*-β compressible plasma. We drive fast-mode turbulence via isotropic wave-vector injection, which ensures a homogeneous forcing mechanism that minimizes the contamination by other MHD modes. We analyze the energy spectrum and structure functions to characterize the energy distribution and isotropy across scales. We find that non-linear steepening causes a Kolmogorov-like ($k^{-1.7}$) fast mode energy spectrum in 3D turbulence, which otherwise exhibits an IK ($k^{-3/2}$) scaling. Further, simulations with varying injection scales are generated to study the dependence of the cascade on the injection scale. We examine the cascade rate by tracking the nonlinear energy transfer through temporal decay of turbulent energy at the injection and inertial scales. Our results are tested against the expected scaling behavior of weak acoustic cascade, which is typically proposed for fast modes in high-β regimes. We find that in the injection scale of turbulence as well as in the inertial range, the cascade rate shows a $k^{-1/2}$ scaling, suggesting that fast modes also follow a weak acoustic cascade in highly compressible *low*-β plasmas, despite of the nonlinear steepening.

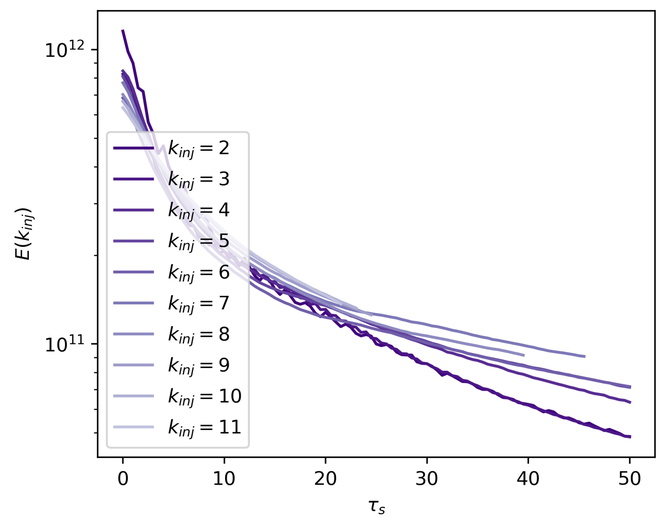
Method

To study fast mode spectrum and cascade independently from other MHD modes, the initial fluctuations in the simulations are constructed to explicitly drive fast magnetosonic waves. The component of the plasma velocity perpendicular to the local mean-field direcrtion is perturbed to generate polarized waves with randomized phases. The parallel component then follows from the fast mode momentum equation. The linearized induction and continuity equations give us the magnetic field and density perturbations, repectively, while the EOS sets the plasma pressure. The plasma frequency satisfies the fast mode dispersion relation,

$$\omega^2 = \frac{1}{2}k^2(c_s^2 + v_A^2)\left[1 + \sqrt{1 - \frac{4k_\parallel^2}{k^2}\frac{v_A^2 c_s^2}{(v_A^2 + c_s^2)^2}}\right]$$

ensuring minimal contamination from Alfvén and slow modes.

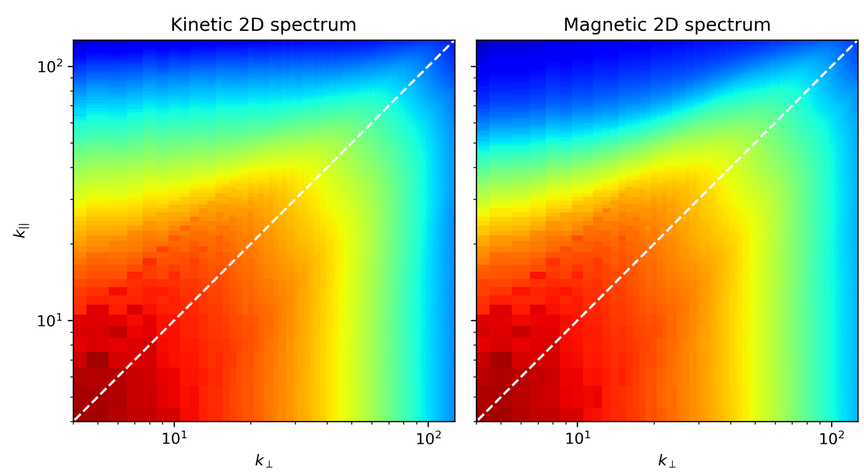
Simulations



Time evolution of the energy at the injection scale in the different simuations.

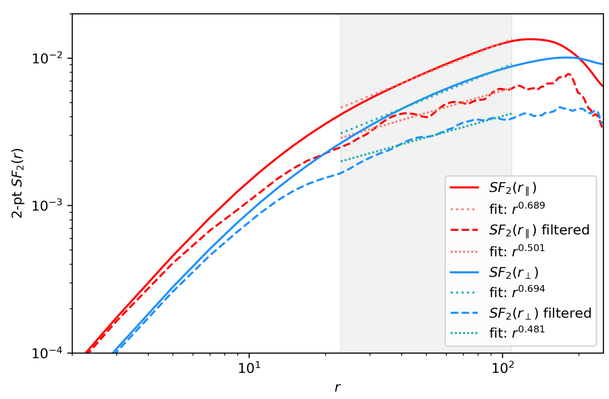
The simulations are performed using the PLUTO code [2]. The conservative MHD equations are solved for non-resistive, non-viscous plasma with numerical dissipation, and an isothermal EOS ($\gamma = 1$). The finite volume HLLD Riemann solver is used to resolve discontinuities, along with 3rd order WENO spatial reconstruction and 3rd order Runge-Kutta time-stepping. The background magnetic field strength ($B_0 \sim v_A$) and the isothermal sound speed (c_s) are controlled to achieve a plasma-$\beta = 0.1$. The box size is unity, and the global mean B-field is fixed along the z direction. Turbulence is driven by injecting fast magnetosonic waves as perturbations in velocity, density and the B-field at $k_{inj} \in [2, 11]$. The amplitude of perturbations is set so that the total energy at the injection scale is consistent across all simulations.

Energy spectrum and structure function



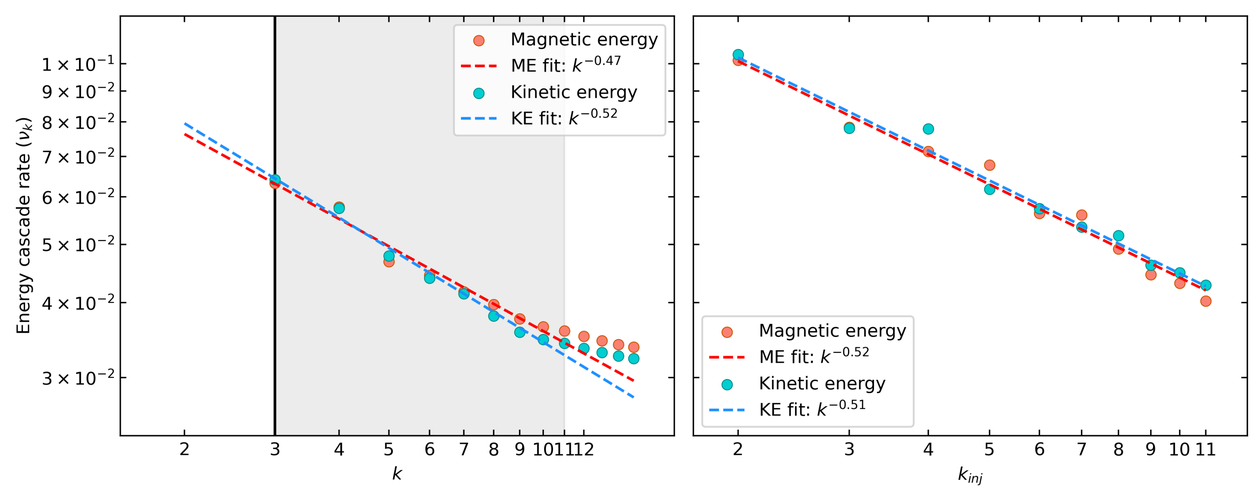
Time averaged 2D kinetic and magnetic energy spectra of fast modes with $k_{\rm inj} = 3$, showing the isotropic energy cascade in the inertial range.

The structure function analysis reveals the impact of nonlinear steepening on fast mode turbulence. The structure functions exhibit slopes of $\sim r^{0.7}$ in both parallel and perpendicular directions, corresponding to a power spectrum slope of $\sim k^{-1.7}$. We find that this steepening arises from sharp velocity and magnetic field gradients that develop from nonlinear steepening during turbulent evolution. When these large gradients are filtered out, the structure function slopes reduce to $\sim r^{1/2}$, yielding the expected $k^{-3/2}$ spectral scaling characteristic of fast magnetosonic modes. Importantly, we find that the a $k^{-1.7}$ spectrum only appears in 3D fast mode simulations, in contrast with the 1D case where a k^{-2} slope typical of shock-dominated regimes is observed [1]. This suggests that 3D effects partially suppress the formation of strong discontinuities, allowing the underlying fast mode cascade dynamics to remain visible.



Second order 2-point structure function of the fast mode velocity in the global parallel and perpendicular directions. The dashed lines show the SF after filtering out large gradients.

Fast mode cascade rate



Wavenumber scaling of the cascade rate of fast mode magnetic and kinetic energy, which obeys the expected scaling behavior of weak acoustic cascade $\nu_k \propto k^{-1/2}$. *Left:* Cascade rate in the turbulence inertial range shown by the shaded region. The vertical line shows the injection scale for the simulation. *Right:* Cascade rate at k_{inj} for simulations with varying injection scales.

Conclusions

- 3D fast mode turbulence in low-β plasmas exhibits an IK-like $k^{-3/2}$ scaling in the inertial range, usually associated with acoustic modes.
- Sharp gradients in velocity and magnetic fields lead to a steeper $\sim k^{-1.7}$ scaling in the energy spectrum, indicating nonlinear steepening.
- The fast mode energy cascade rate follows the expected weak acoustic cascade behavior despite the nonlinear steepening, with a $k^{-1/2}$ scaling at the injection and inertial scales. This suggests that nonlinear steepening does not significantly alter the cascade dynamics.

Acknowledgements

The authors gratefully acknowledge the computing time made available to them on the high-performance computer "Lise" at the NHR center NHR@ZIB. This center is jointly supported by the Federal Ministry of Education and Research and the state governments participating in the NHR (www.nhr-verein.de).

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